What do you see here?

https://commons.wikimedia.org/wiki/File:Nightingale-mortality.jpg#/media/File:Nightingale-mortality.jpg

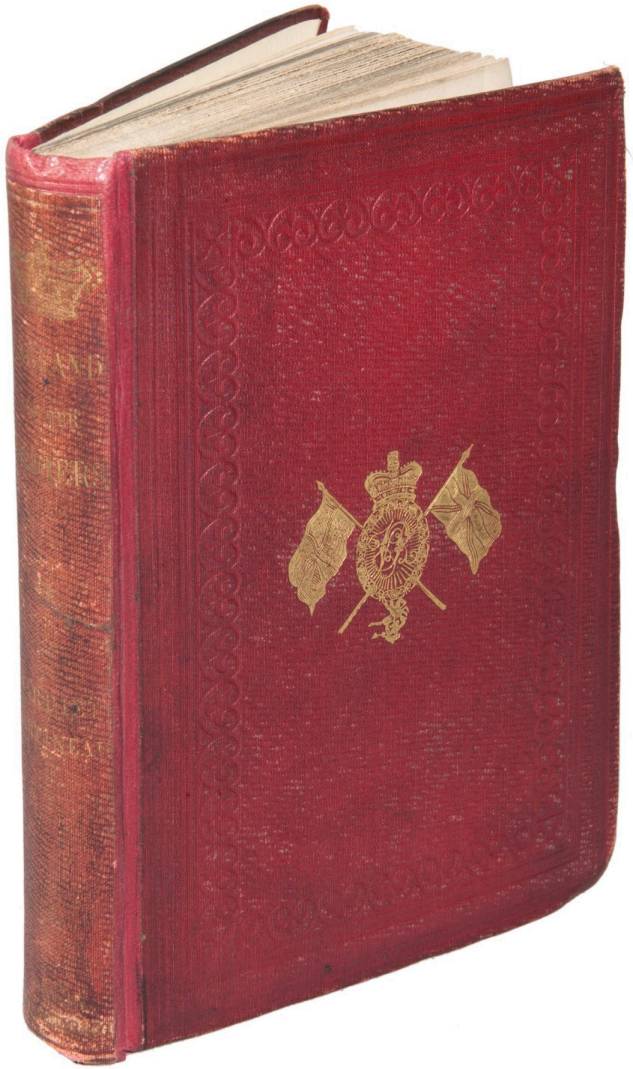


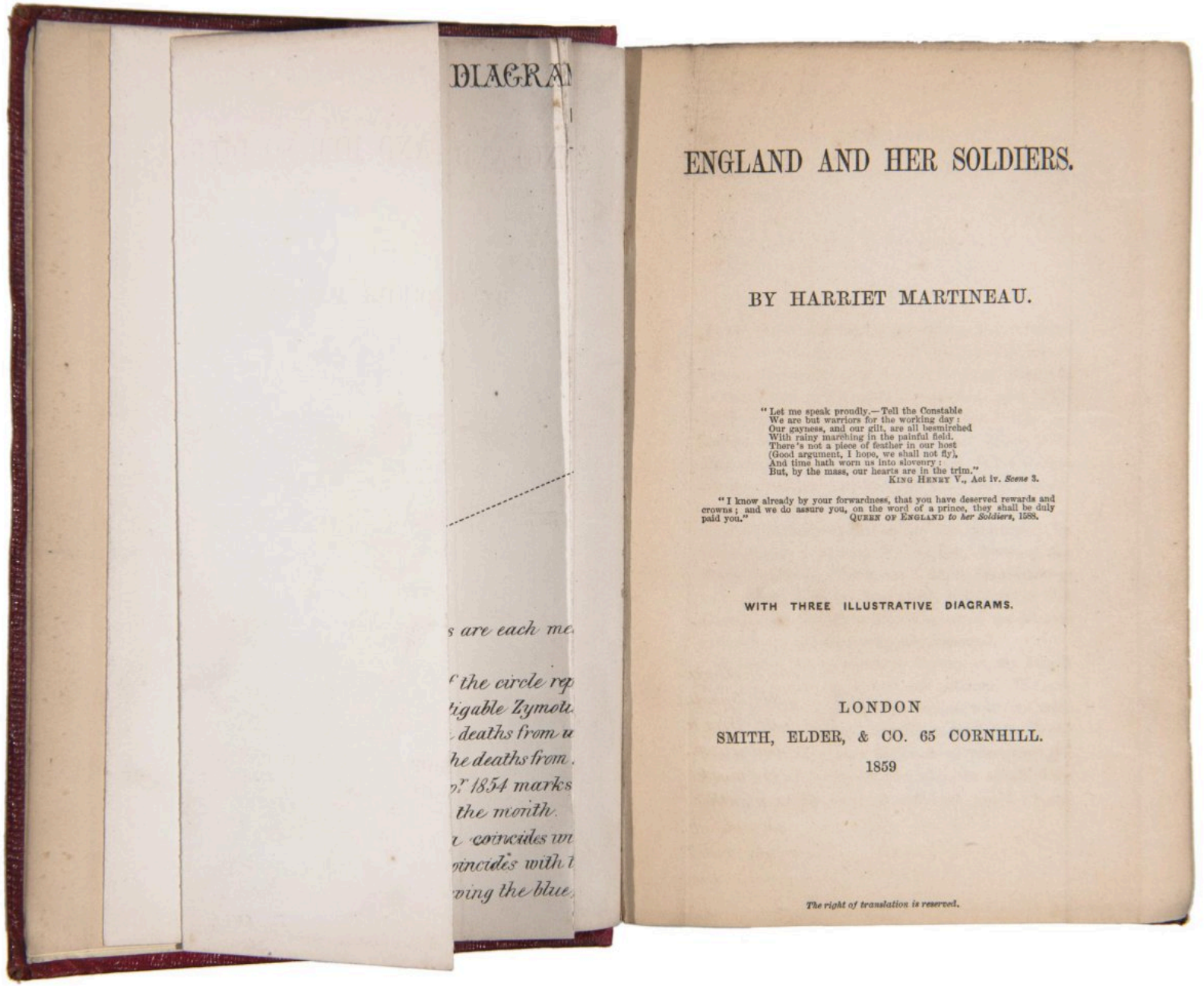
The lady with the lamp

Florence Nightingale (1820-1910)



MISS NIGHTINGALE, IN THE HOSPITAL, AT SCUTARI.—(SEE PRECEDING PAGE.)





ENGLAND AND HER SOLDIERS.

BY HARRIET MARTINEAU.

"Let me speak proudly.—Tell the Constable
We are but warriors for the working day:
Our gayness, and our gilt, are all besmirched
With rainy marching in the painful field.
There's not a piece of feather in our host
(Good argument, I hope, we shall not fly),
And time hath worn us into slovenry:
But, by the mass, our hearts are in the trim."
KING HENRY V., Act iv. *Scene* 3.

"I know already by your forwardness, that you have deserved rewards and crowns; and we do assure you, on the word of a prince, they shall be duly paid you."
QUEEN OF ENGLAND *to her Soldiers*, 1588.

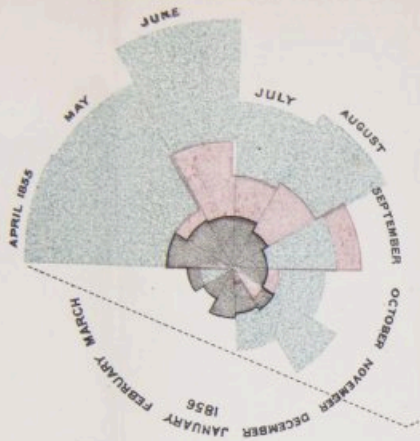
WITH THREE ILLUSTRATIVE DIAGRAMS.

LONDON
SMITH, ELDER, & CO. 65 CORNHILL.
1859

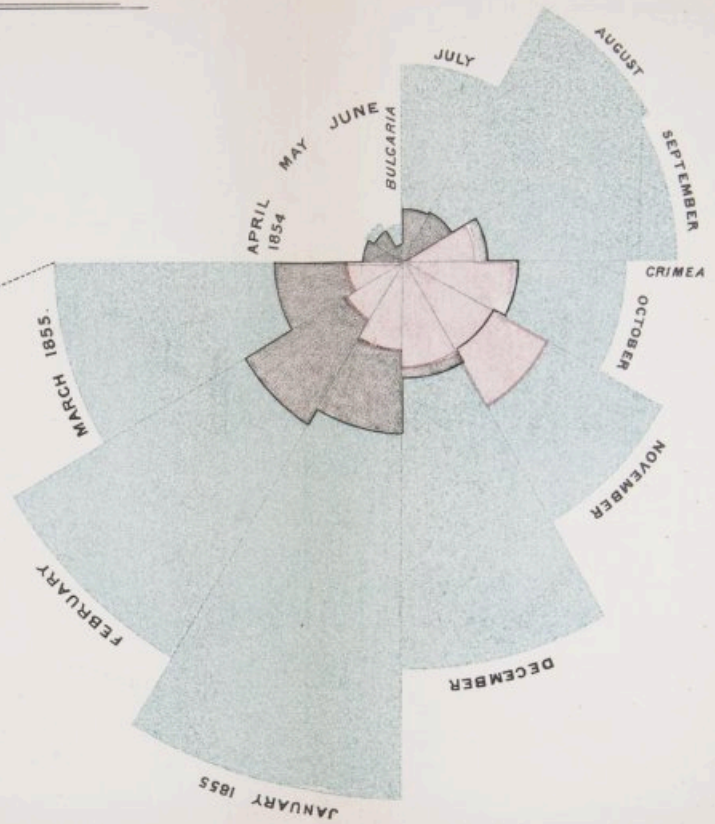
The right of translation is reserved.

DIAGRAM OF THE CAUSES OF MORTALITY IN THE ARMY IN THE EAST.

2.
APRIL 1855 TO MARCH 1856.



1.
APRIL 1854 TO MARCH 1855.



The Areas of the blue, red, & black wedges are each measured from the centre as the common vertex.

The blue wedges measured from the centre of the circle represent area for area the deaths from Preventible or Mitigable Zymotic diseases, the red wedges measured from the centre the deaths from wounds, & the black wedges measured from the centre the deaths from all other causes.

The black line across the red triangle in Nov.r 1854 marks the boundary of the deaths from all other causes during the month.

In October 1854, & April 1855, the black area coincides with the red, in January & February 1855, the blue coincides with the black.

The entire areas may be compared by following the blue, the red & the black lines enclosing them.

ENGLAND AND HER SOLDIERS.

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WITH THREE ILLUSTRATIVE DIAGRAMS.

LONDON
SMITH, ELDER, & CO. 65 CORNHILL.
1859

The right of translation is reserved.

Why Python?


matplotlib


seaborn


plotly

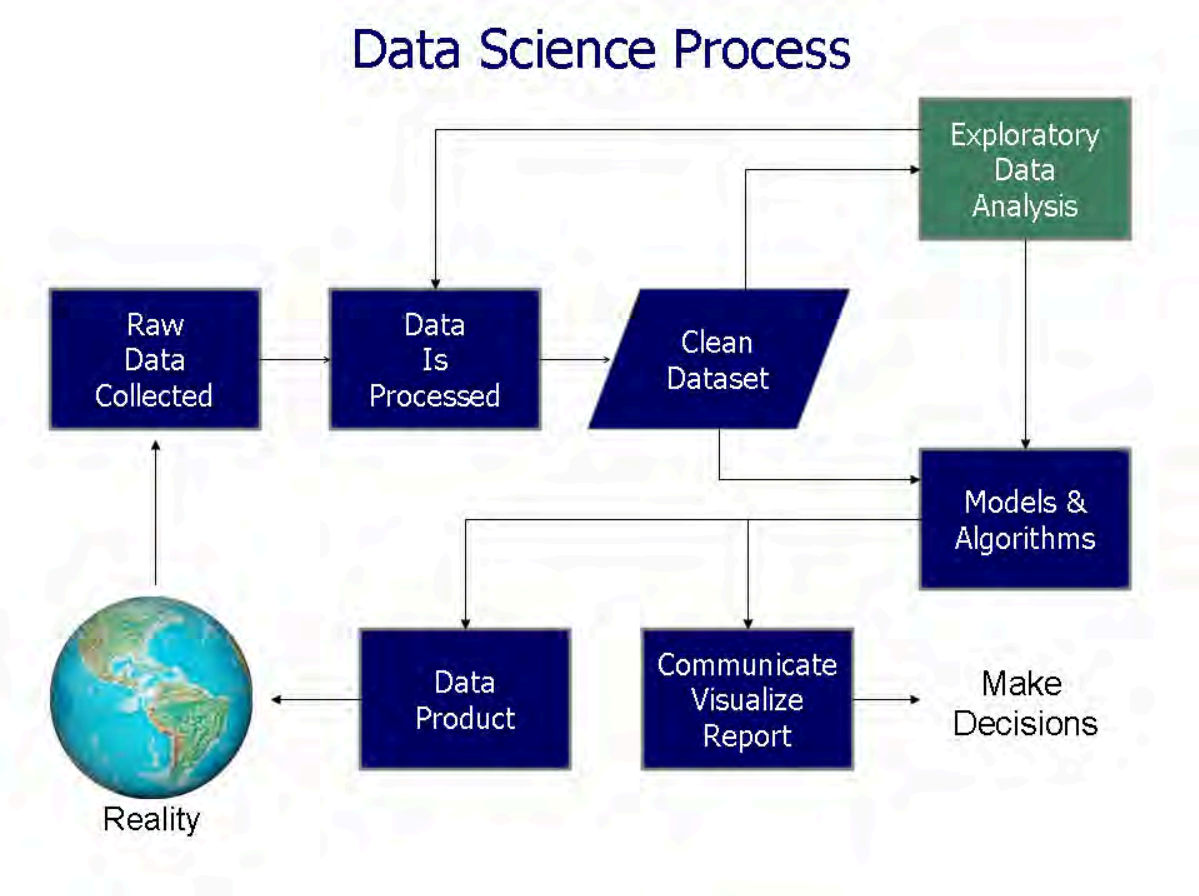

bokeh

“The greatest value of a picture is when it forces us to notice what we never expected to see.”



— John Tukey, 1977

EDA



Exploratory Data Analysis

① Discovery through Visualization

Seeing patterns, trends, and outliers that numbers alone cannot show.

可視化による発見

数字だけでは見えないパターンや傾向、外れ値を「見る」ことで発見する。

② Iterative Exploration

Analyze, visualize, question, and refine repeatedly to uncover deeper insights.

反復的な探究

可視化と分析を繰り返し行い、データの奥にある意味を掘り下げる。

③ Openness to Surprise

Expect the unexpected — EDA values curiosity over confirmation.

意外性への開放性

仮説の検証よりも、好奇心と「予想外の発見」を重視する。

datasaurus.csv



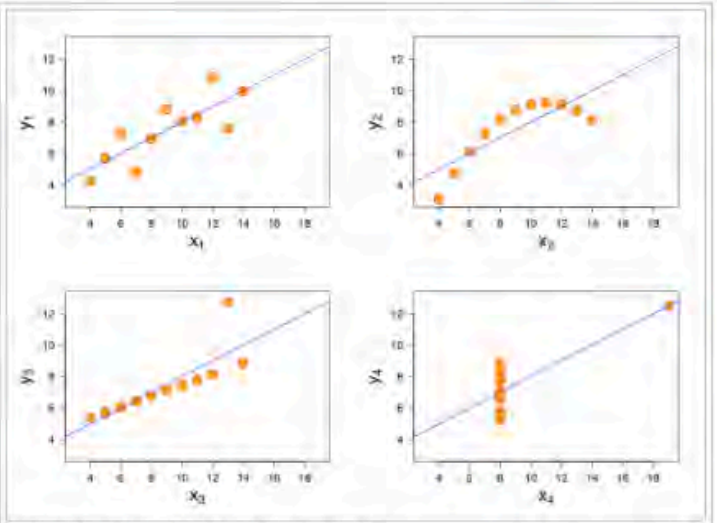
Find the dinosaur!

Anscombe's quartet

概要 [編集]

アンスコムの例は右のグラフに示された4つのデータセットからなる。4つ全てについて以下が成り立つ。

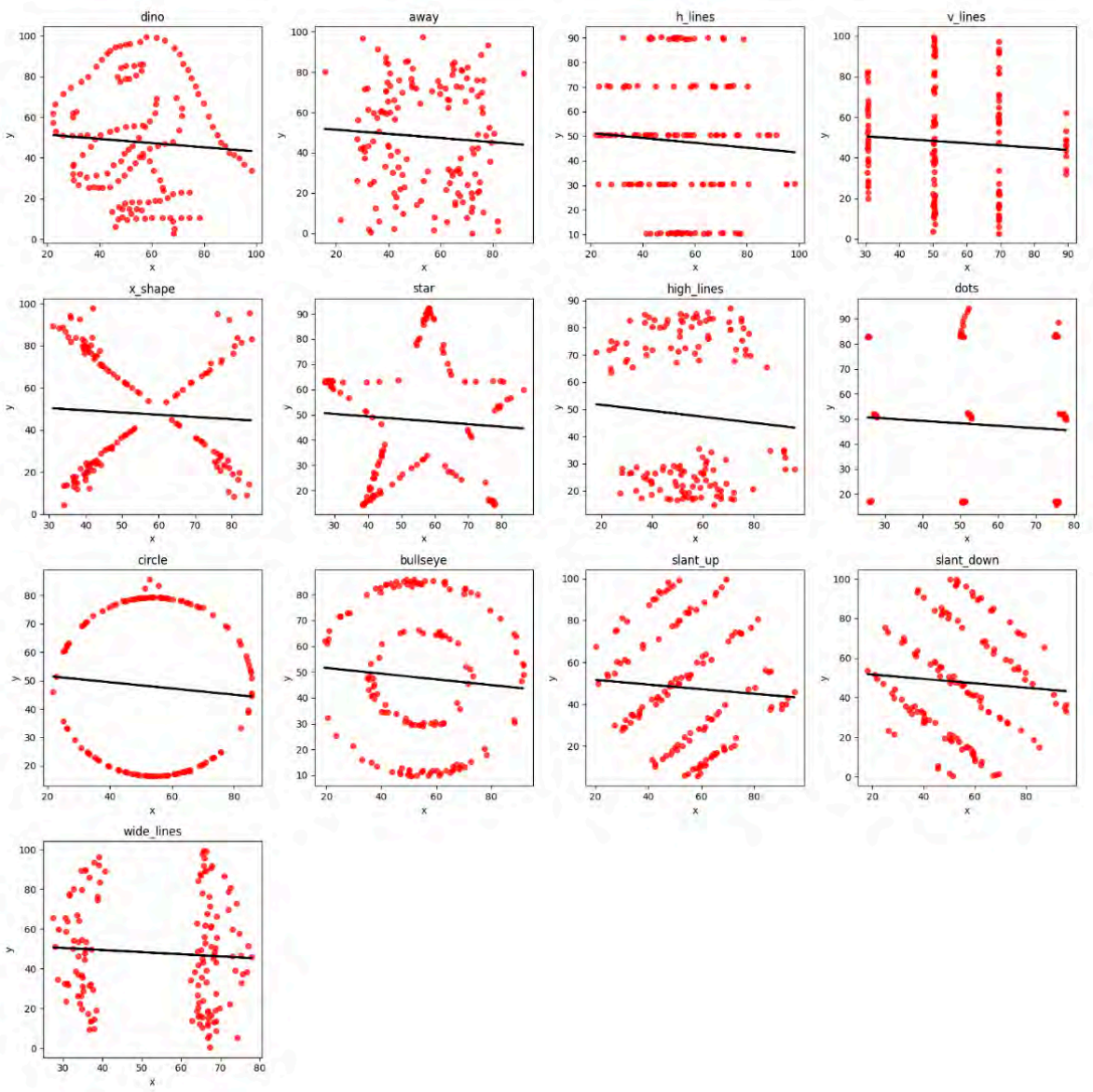
統計量	値
x の平均	9 (正確に一致)
x の標本分散	11 (正確に一致)
y の平均	7.50 (小数第2位まで一致)
y の標本分散	4.122 or 4.127 (小数第3位まで一致)
x と y の相関係数	0.816 (小数第3位まで一致)
回帰直線	$y = 3.00 + 0.500x$ (それぞれ小数第2位、小数第3位まで一致)

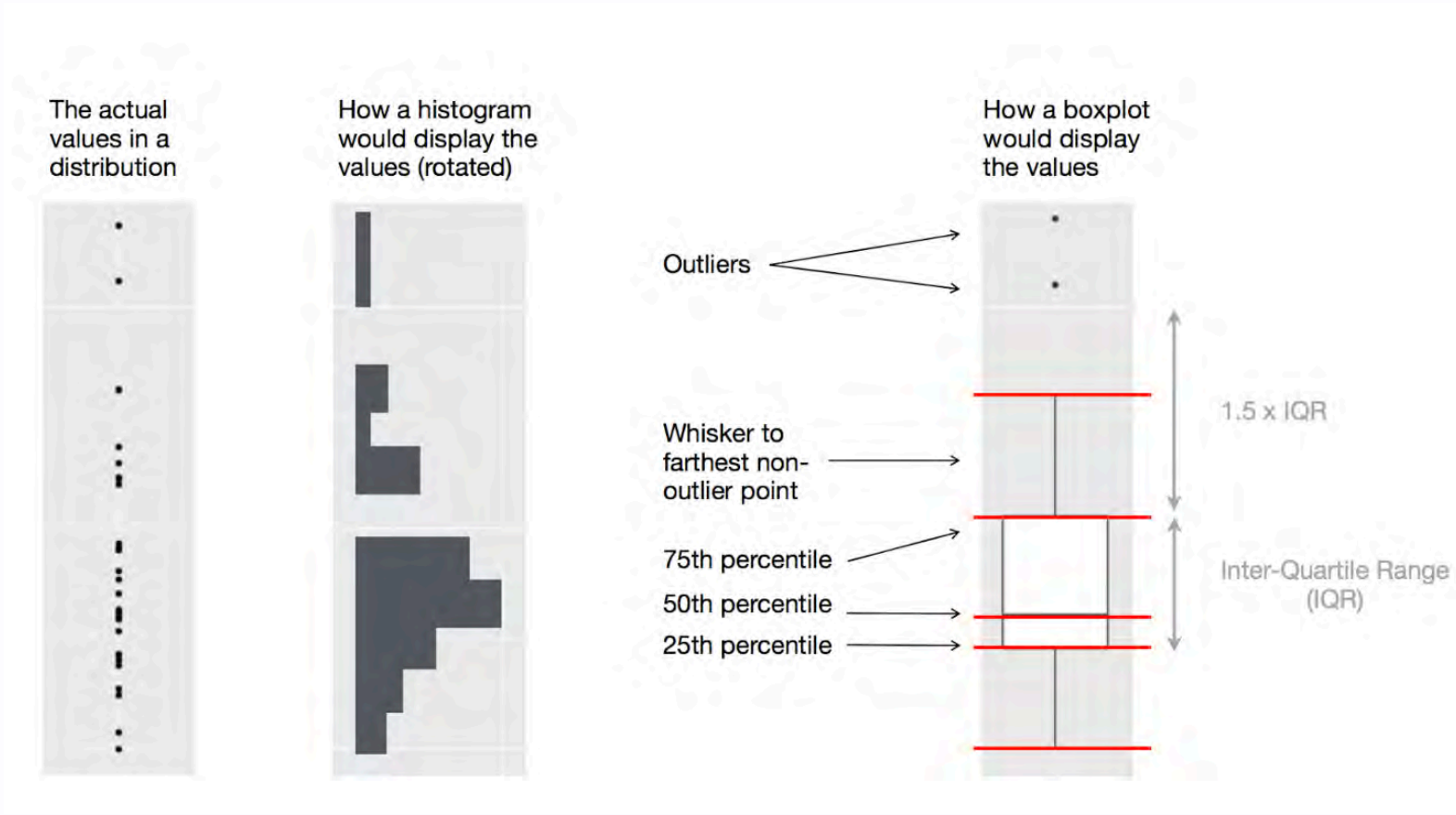


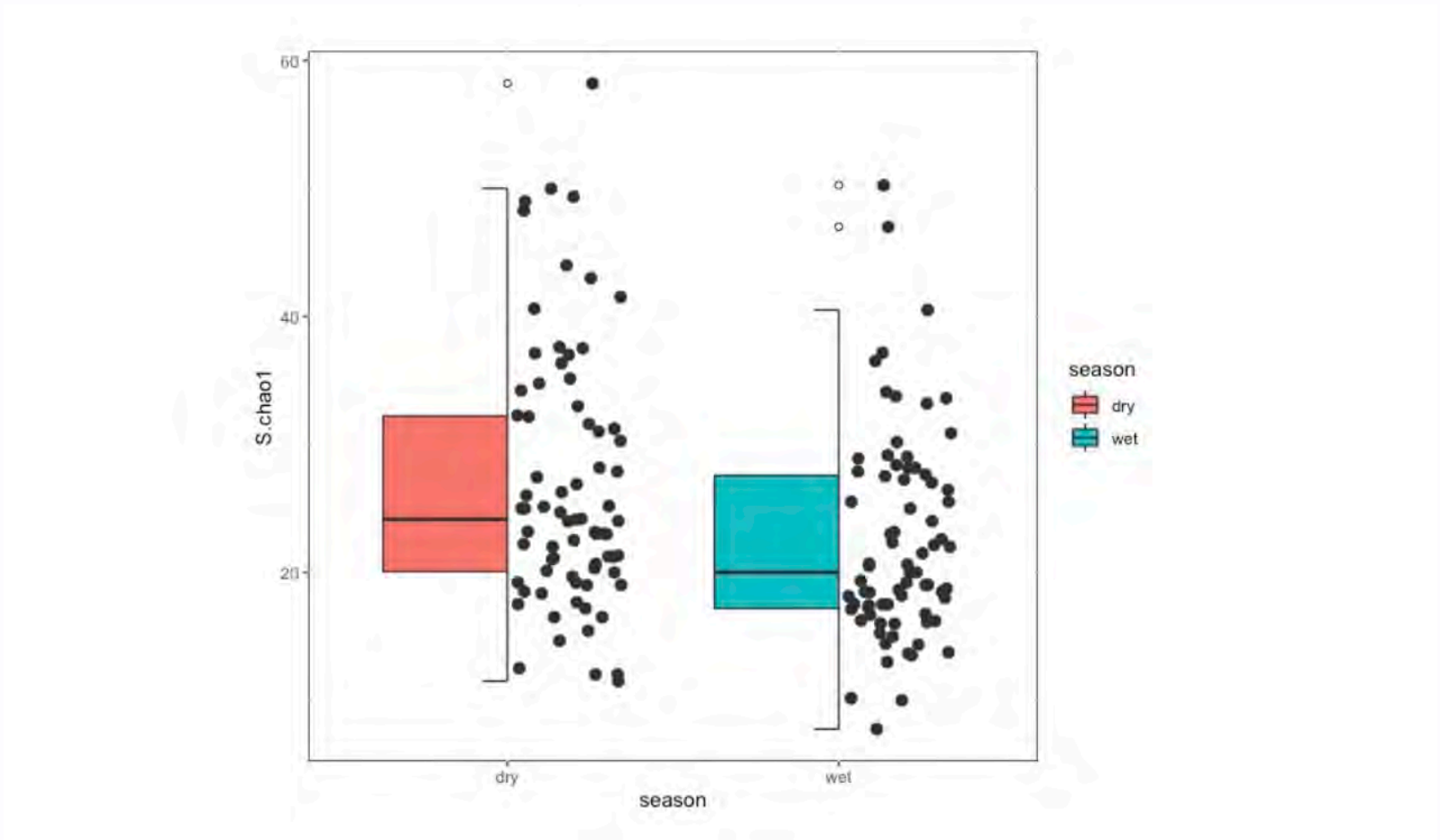
回帰直線が同じになる4つの散布図

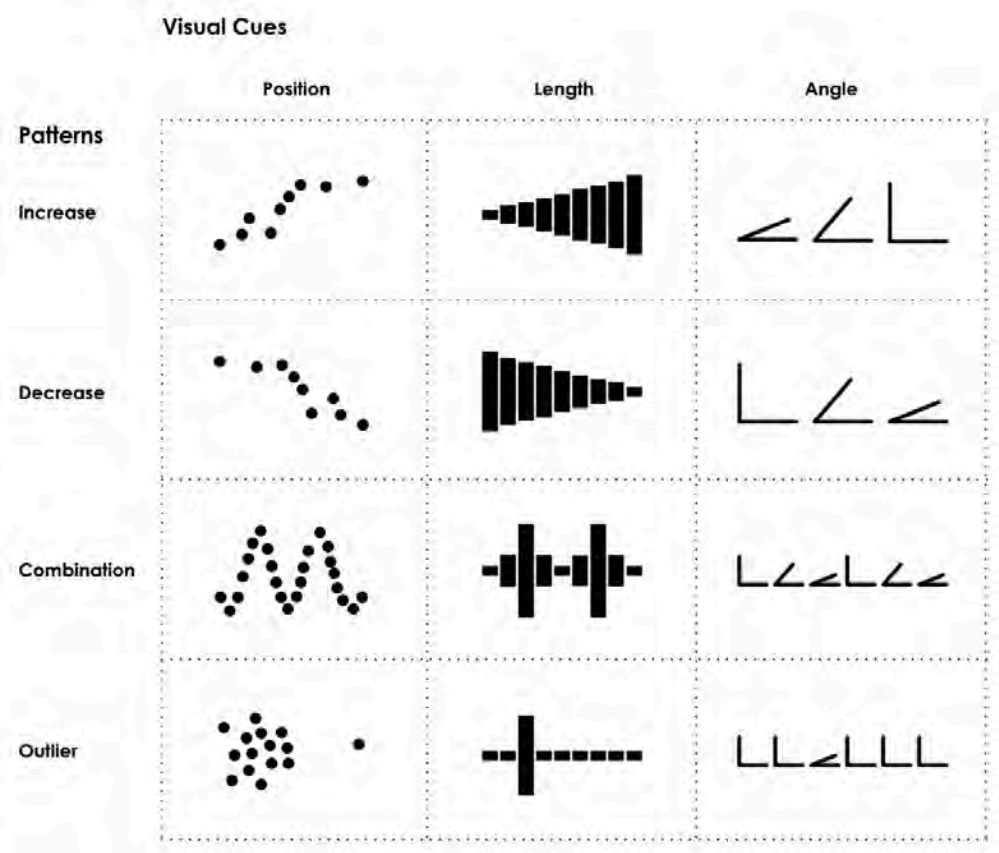
https://ja.wikipedia.org/wiki/アンスコムの例

In Python


dino
away
h_lines
v_lines
x_shape
star
high_lines
dots
circle
bullseye
slant_up
slant_down
wide_lines
x
y


The actual values in a distribution
How a histogram would display the values (rotated)
How a boxplot would display the values
Outliers
Whisker to farthest non-outlier point
75th percentile
50th percentile
25th percentile
1.5 x IQR
Inter-Quartile Range (IQR)


60
40
20
S.chao1
dry
wet
season
season
dry
wet



https://www.nature.com/articles/497186a

Geminiとは？

Google DeepMind / December 2023

言葉・画像・コードを扱う、GoogleのAIモデル群

名はふたご座の双子、カストルとポルックスから。

BrainとDeepMindの合流も、この名の由来。

Google: Geminiの名前の由来

Vibe coding

Andrej Karpathy / February 2025

やりたいことを、ふつうの言葉で伝える。

AIがコードを書き、実行結果を見て対話を続ける。

今日はPythonでデータを探索します。

Karpathyの言葉（2025年2月、Simon Willisonによる引用）

Google Colabの準備

Python runs in your browser

1　ログインして、新しいノートブックを作る。

2　必要なら「接続」。左のフォルダアイコンを開く。

3　「セッション ストレージにアップロード」を選ぶ。

4　datasaurus.csvを選び、一覧への表示を待つ。

Google Colabを開く

最初のひと言

Geminiのきらきらマークをクリック。右のパネルで。

データを読み込んで

Load the data.

ここからは、自分で探索。恐竜を見つけよう。

Geminiが使えない場合は、教員に相談してください。

Geminiと、試してみる

Conversation + Python + output

Geminiがコードセルを作ったら、提案を確認。

「Accept and run」をクリックして実行。

結果を見て、次に知りたいことを自分で考える。

エラーも会話の続き。直して、また試す。

Google: ColabとGeminiの使い方

動いた。では、正しい？

You decide whether the result makes sense

図は、アップロードしたデータを表している？

軸・数値・説明は、元のデータと合っている？

知らないコードの働きは、Geminiに説明してもらう。

個人情報は入れず、ノートブックをDriveに保存。

Google Colab FAQ: AIの利用とデータの扱い

datasaurus.csv

1,846 points / 13 datasets

dataset	x	y
dino	55.3846	97.1795
dino	51.5385	96.0256
dino	46.1538	94.4872
dino	42.8205	91.4103
dino	40.7692	88.3333
dino	38.7179	84.8718

Preview data / Download CSV

dataset = group name x, y = coordinates of one point